3.4.3 Earth's atmosphere

Students should have knowledge and understanding of the following content.

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Outcome 5			
During the first billion years of the Earth's existence, there was intense volcanic activity that released gases that formed the early atmosphere and water vapour that condensed to form the oceans. The early atmosphere was mainly carbon dioxide with little or no oxygen.		5.9.1.2	4.4.1.1

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
From about three billion years ago, algae and plants developed and produced the oxygen that is now in the atmosphere, by a process called photosynthesis. Photosynthesis can be represented by the word equation: carbon dioxide + water → glucose + oxygen	**Suggested activity for TDA** Investigate the production of oxygen by aquatic plants in different conditions by counting bubbles.	5.9.1.3	4.4.1.1
Outcome 6			
Carbon dioxide was removed from the early atmosphere by dissolving in the oceans and by photosynthesis. Most of the carbon from the carbon dioxide gradually became locked up in rocks as carbonates and fossil fuels.		5.9.1.4	4.4.1.1
The Earth's atmosphere is now about four-fifths (80%) nitrogen and about one-fifth (20%) oxygen, with small amounts of other gases, including carbon dioxide, water vapour and argon, which is a noble gas.	**Suggested activity for TDA** Compare the amount of carbon dioxide in fresh air and exhaled air.	5.9.1.1	